

TAFWELA

Functional Requirements

(DPEI) | 2026

1. Purpose

This document lists all functional requirements for the TAFWELA application. Each requirement defines a specific system behavior or capability. Requirements are organized by module and include priority, responsible actor, related user story, and implementation status.

2. Requirements Summary

Module	Total	Must Have	Should Have	Could Have	Owner
Authentication	5	5	0	0	Youssef / Mohannad
Map & GPS	7	6	0	1	Ziad
Station Details	4	3	0	1	Mohamed
Search & Filter	4	0	4	0	Mohamed
Favourites	4	0	4	0	Mohamed
Onboarding	3	0	3	0	Youssef
Staff Panel	6	3	3	0	Omar
Real-time Sync	2	2	0	0	Mohannad
TOTAL	35	19	14	2	

3. Functional Requirements Register

3.1 Authentication

ID	Requirement Description	Priority	Actor	Related US	Status		
FR-01	The system shall allow drivers to register using email and password via Firebase Auth.	Must Have	Driver	US-10	Planned		
FR-02	The system shall allow station staff to log in using pre-created authorized staff accounts.	Must Have	Staff	US-13	Planned		
FR-03	The system shall validate email format and enforce a minimum password length of 8 characters.	Must Have	Driver / Staff	US-10	Planned		
FR-04	The system shall redirect users to the correct screen based on their role (Driver → Map, Staff → Dashboard).	Must Have	Driver / Staff	US-10, US-13	Planned		
FR-05	The system shall display appropriate error messages for invalid login credentials.	Must Have	Driver / Staff	US-10	Planned		

3.2 Map & GPS

ID	Requirement Description	Priority	Actor	Related US	Status		
FR-06	The system shall display an interactive Google Map centered on the driver's current GPS location.	Must Have	Driver	US-01	Planned		
FR-07	The system shall automatically detect the driver's GPS location using the Geolocator package.	Must Have	Driver	US-03	Planned		
FR-08	The system shall display fuel stations as custom markers on the map within a configurable radius.	Must Have	Driver	US-01	Planned		
FR-09	The system shall color-code each marker based on congestion status: Green (Low), Yellow (Medium), Red (High), Gray (No Fuel).	Must Have	Driver	US-02	Planned		
FR-10	The system shall calculate and display the distance between the driver and each station.	Must Have	Driver	US-01	Planned		
FR-11	The system shall handle GPS permission denial gracefully by displaying a manual location prompt.	Must Have	Driver	US-03	Planned		
FR-12	The system shall support marker clustering when the map is zoomed out.	Could Have	Driver	US-01	Planned		

3.3 Station Details

ID	Requirement Description	Priority	Actor	Related US	Status		
FR-13	The system shall display a Station Details screen when a driver taps a marker, showing: name, address, distance, congestion status, and working hours.	Must Have	Driver	US-04	Planned		
FR-14	The system shall display a color-coded congestion badge on the Station Details screen.	Must Have	Driver	US-04	Planned		
FR-15	The system shall provide a Navigate button that launches Google Maps with the station as the destination.	Must Have	Driver	US-05	Planned		
FR-16	The system shall provide a Share button to share station name and location via the device share sheet.	Could Have	Driver	US-11	Planned		

3.4 Search & Filter

ID	Requirement Description	Priority	Actor	Related US	Status		
FR-17	The system shall provide a search bar allowing drivers to search stations by name or area.	Should Have	Driver	US-06	Planned		
FR-18	The system shall filter displayed stations in real time as the driver types in the search bar.	Should Have	Driver	US-06	Planned		
FR-19	The system shall allow drivers to filter stations by congestion status (e.g., Green only).	Should Have	Driver	US-07	Planned		
FR-20	The system shall provide a List View as an alternative to the Map View, displaying stations sorted by distance.	Should Have	Driver	US-08	Planned		

3.5 Favourites

ID	Requirement Description	Priority	Actor	Related US	Status		
FR-21	The system shall allow drivers to save stations as favourites by tapping a heart icon.	Should Have	Driver	US-09	Planned		
FR-22	The system shall store favourites locally using shared_preferences.	Should Have	Driver	US-09	Planned		
FR-23	The system shall display a Favourites screen showing all saved stations with their current congestion status.	Should Have	Driver	US-09	Planned		
FR-24	The system shall allow drivers to remove a station from favourites.	Should Have	Driver	US-09	Planned		

3.6 Onboarding

ID	Requirement Description	Priority	Actor	Related US	Status		
FR-25	The system shall display 3–4 onboarding screens on first launch explaining the app's key features.	Should Have	Driver	US-12	Planned		
FR-26	The system shall allow users to skip the onboarding flow.	Should Have	Driver	US-12	Planned		
FR-27	The system shall not show the onboarding screens again after the first launch.	Should Have	Driver	US-12	Planned		

3.7 Staff Panel

ID	Requirement Description	Priority	Actor	Related US	Status		
FR-28	The system shall provide a dedicated login screen for station staff.	Must Have	Staff	US-13	Planned		
FR-29	The system shall display a Staff Dashboard showing the station name, current congestion status, and last update timestamp.	Must Have	Staff	US-14, US-17	Planned		
FR-30	The system shall allow staff to update congestion status with four options: Low / Medium / High / No Fuel.	Must Have	Staff	US-14	Planned		
FR-31	The system shall display a confirmation dialog before saving a congestion status update.	Should Have	Staff	US-15	Planned		
FR-32	The system shall record each status change with the staff member's ID and timestamp in Firestore.	Should Have	Staff	US-16	Planned		
FR-33	The system shall display a change history log showing the last 10 status updates for the station.	Should Have	Staff	US-16	Planned		

3.8 Real-time Sync

ID	Requirement Description	Priority	Actor	Related US	Status		
FR-34	The system shall use Firestore real-time listeners to update station marker colors within 1 second of a staff update.	Must Have	Driver	US-02	Planned		
FR-35	The system shall detach Firestore listeners when the screen is disposed to prevent memory leaks.	Must Have	Driver / Staff	—	Planned		

